

Lecture 10 -

2026-05-10.

I. Overview

In the last lecture, we began the variational derivation of Maxwell's equations by constructing the total action for point charges plus the electromagnetic field, incorporating free-particle, interaction, and quadratic free-field terms dictated by Lorentz invariance, gauge invariance, and the superposition principle. We evaluated the first variation of the free-field action with respect to the four-potential, applying spacetime integration by parts to isolate the dynamic field term. Point charges were represented via Dirac delta functions to define the charge density, current density, and the four-current $J^\mu = (c\rho, \mathbf{J})$, which was proven to be a proper Lorentz four-vector obeying the covariant continuity equation $\partial_\mu J^\mu = 0$ for local charge conservation. Finally, combining the free-field variation with the interaction term yields the covariant inhomogeneous Maxwell equations, $\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu$.

We start from the four-current

$$J^{\mu(x,t)} = \begin{pmatrix} c\rho \\ \vec{J} \end{pmatrix} = \sum_n e_n \delta(\vec{x} - \vec{x}_n(t)) \frac{dx_n^\mu}{dt} \quad (1)$$

where $x_n^0 = ct$

By using the four-current we can write S_{int} by the following:

$$S_{\text{int}} = \sum \frac{e_n}{c} \int dt A_\mu \frac{dx_n^\mu}{dt} = -\frac{e_n}{c} \int dt \int d^3x \delta(\vec{x} - \vec{x}_n) d\frac{x_n^\mu}{dt} \cdot A_\mu(\vec{x}, t) \quad (2)$$

By utilizing the delta function we able to "remove" the integral and only take the values where $\vec{x} = \vec{x}_x$. (תכונת הסינון של פונקציית דלטה) Cause A_μ is at the particle location.

$$S_{\text{int}} = -\frac{1}{c^2} \int d^4x J^\mu(\vec{x}, t) A_\mu(\vec{x}, t) \quad (3)$$

In particular, the variation of S_{int} is:

$$\delta S_{\text{int}} = -\frac{1}{c^2} \int d^4x J^\mu(\vec{x}, t) \delta A_\mu(\vec{x}, t) \quad (4)$$

We have variation for S_{int} and the variation for S_{field} then we have the variation for the entire field

By the minimal action principal we acquire that this expression is zero

$$\delta S = \frac{1}{4\pi c} \left[\partial_\mu F^{\mu\nu} - \frac{4\pi}{c} J^\nu \right] A_{\nu(\vec{x}, t)} \stackrel{!}{=} 0 \Rightarrow \partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu \quad (5)$$

And this is maxwell's equations in compact form by using the E.M field tensor

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu \quad (6)$$

This is the field part. The particle part is

$$m \frac{du_\mu}{d\tau} = \frac{e}{c} F_{\mu\nu} u^\nu \quad (7)$$

Those are *maxwell's equations*.

All of the electrodynamics are packed in these compact equation.

We were able to derive this only from

- Minimal action principal
- Symmetry arguments
- Experimentals samples

When we have a good understanding of the system the equations could become very simple.

Lets try to solve this equations. We have 4 equations, For $\nu = 0$:

$$\partial_{\mu 0} = \frac{4\pi}{c} J^0 \quad (8)$$

For $\nu = 0$ we will get only 0 components (from the E.M field) For the spacial indicis we will get

$$\partial_i F^{i0} = 4\pi\rho \Rightarrow \operatorname{div} \mathbf{E} \quad (9)$$

Note that $F^{i0} = -F_{i0}$ and $A^\mu = (\varphi, \mathbf{A})$ hence

$$F^{i0} = -F_{i0} = \underbrace{-\partial_i A_0}_{-\nabla\varphi} + \underbrace{\partial_0 A_i}_{-\frac{1}{c}\frac{d\mathbf{A}}{dt}} = \mathbf{E}_i \quad (10)$$

Thus,

$$F^{i0} = \nabla\varphi + \partial_0 A_i - \frac{1}{c} \frac{d\mathbf{A}}{dt} = \mathbf{E} = 4\pi\rho \quad (11)$$

Lets demonstartate it by taking $\mu = 1$

$$\begin{aligned} \partial_\mu F^{\mu 1} &= \frac{4\pi}{c} J^1 = \partial_0 F^{01} + \partial_2 F^{21} + \partial_3 F^{31} = \frac{4\pi}{c} J^1 \\ F^{01} &= -E_x, J^1 = J_x \\ F^{21} = F_{21} &= \partial_2 A_1 - \partial_1 A_2 = -\frac{\partial A_x}{\partial_y} + \frac{\partial A_y}{\partial_x} = \operatorname{rot} \mathbf{A}_z = B_z \\ F^{31} &= -B_y \end{aligned} \quad (12)$$

Note: Magneic monopoles are theoretic and can symmetrize maxwell's equations.